

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

All traffic from R3 to 192.168.2.0/24 will pass to R1.

The total cost of 129 comes from the paths the traffic has to take.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

64(T1 serial link)+1(R2 Gigabit 0/0 LAN link)=65

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

Please ensure reference bandwidth is consistent across all routers.

Why would you want to change the OSPF default reference-bandwidth?

__To make sure the calculate of the the cost for the faster link is correct, if the calculate result is small (0.x), it just return all cost is 1, so we have to make the bandwidth bigger to see what exactly the cost is..

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24's cost=781(R1 S0/0/1)+1(R3 G0/0)=782

192.168.23.0/30's cost=781(R1 S0/0/1)+64(R3 S0/0/1)=845

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The route to the 192.168.23.0/24 network travels over two serial links, and each link need 781 cost. 781 + 781 = 1,562

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

R2 path cost is 1563 and direct go R3 path cost is 1566 ,_because OSPF will choose a path that using the least cost, so choosed R2 path.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Router ID assignments help decide which router is which one. If the router ID is linked to an active interface, it can change if that interface stops working or have some error.

To prevent this, it's better to use router ID assignment to make the ID will not change.

2. Why is the DR/BDR election process not a concern in this lab?

In this lab, we are using serial links that connect directly to each other, and DR/BDR election is important only in networks with many connections so DR/BDR election process not a concern in this lab.

3. Why would you want to set an OSPF interface to passive?

This is because this can stops it from sharing extra OSPF routing information, which saves bandwidth.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

Using Areas help manage a big network by splitting it into smaller network. OSPF make sure the Link-State Advertisements will not spread too much and keep the Link-State Database smaller in each router, which make the work of router lower and make the network more easy to go.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

__ network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

EIGRP will be used for routing decision, because router check Administrative Distance first before looking at other numbers. Since EIGRP has the lowest AD, it's considered the most "trustworthy".